

CESAR SCHOOL

LUCAS ANDRADE ABRÃO

**EVALUATION OF MUSICAL GENERATION MODELS
IN THE FREVO STYLE USING NEURAL
NETWORKS**

RECIFE

2024

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Dissertation presented to the Master's program in Software Engineering at the Center for Advanced Studies and Systems of Recife - CESAR School, as a requirement for obtaining the title of Master in Software Engineering.

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**RECIFE
2024**

Dedication

This work is totally dedicated to God for his goodness, favor and the wonderful work of the Lord Jesus Christ on the cross.

I love you, God!

I also dedicate it to my parents for all their love, affection and dedication to all their children.

Thank you for everything!

Thanks

I thank God first of all for the gift of life and for having blessed me at every moment of my life, always with His grace, love and protection. Thank you, God!

To my parents, Edir Abrão and Elizabeth Abrão, for all their affection, dedication, good example and honesty, who even in life's greatest difficulties always showed love for God, for their children and were a true reference point for integrity and honesty. Words cannot express the gratitude for everything they have done for all those around them!

To my brothers and sisters for making our family unique and special, even with all the defects, the qualities and love we feel for each other are always greater than any difference.

To my friends who always encouraged me and to my colleagues who helped me along the way.

To my advisor, Professor Dr. André Câmara, for his encouragement, support, direction and professionalism in conducting this research.

CESAR School, for the excellent team of professors, advisors and professionals who make the institution a true center of research and innovation.

Summary

Exploring the convergence between advances in Artificial Intelligence and musical expression, this study aims to investigate the application of advanced Deep Learning techniques in the context of automatic music generation. With an emphasis on models such as LSTM, WaveGAN, CNN, Autoencoders and GRU, the research aims to create audio time series in order to compose music in the Frevo style - an authentic and influential musical genre in the Northeast of Brazil. The database, made up of 30 Frevo songs, will serve as key pieces for training the neural networks. The performance of the networks was evaluated using metrics such as MAE, MSE, RMSE, SNR and PSNR, complemented by spectral metrics, visual evaluation, SPL and SNDR, providing a comprehensive analysis of the quality of the audio time series generated. Finally, the performance of each neural network was analyzed and validated using the same hyperparameters or nearby data based on its architecture, respectively. These analyses showed that some networks generate audio better and closer to reality (such as the GRU network) than others (such as the Autoencoders), based on the values assigned to the hyperparameters in this study. The study highlights not only the capacity of neural networks in the personalized creation of musical content, but also seeks to enrich artistic expressiveness by connecting technological innovations to the rich and unique musical tradition of Frevo. Thus, by helping people to compose music in the Frevo style using the assistance of artificial intelligence, the study points to the importance of identifying the best neural network for this purpose among the neural networks analyzed by the metrics used in this study.

Key words

Audio; deep learning; artificial intelligence; metrics; neural networks.

Abstract

Exploring the convergence between advances in Artificial Intelligence and musical expression, this study aims to investigate the application of advanced Deep Learning techniques in the context of automatic music generation. With an emphasis on models such as LSTM, WaveGAN, CNN, Autoencoders, and GRU, the research seeks to create temporal series of audio with the goal of composing music in the Frevo style - an authentic and influential musical genre in Northeast Brazil. The dataset, consisting of 30 Frevo songs, serves as key pieces for training the neural networks. The performance evaluation of the networks was conducted using metrics such as MAE, MSE, RMSE, SNR, and PSNR, complemented by spectral metrics, visual evaluation, SPL, and SNDR, providing a comprehensive analysis of the quality of the generated audio temporal series. In the end, analyses and validations of the performance of each neural network were made using the same hyperparameters or close data based on their architecture, respectively. These analyses showed that some networks have better audio generation and are closer to reality (such as the GRU network) than others (such as Autoencoders), based on the values assigned to the hyperparameters in this study. The study highlights not only the ability of neural networks in the personalized creation of musical content but also seeks to enrich artistic expressiveness by connecting technological innovations with the rich and unique musical tradition of Frevo. Thus, by helping people compose music in the Frevo style using the assistance of artificial intelligence, the study points to the importance of identifying the best neural network for this purpose among the neural networks analyzed by the metrics used in this study.

Key-

Artificial intelligence; audio; deep learning; metrics; neural networks.

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1 INTRODUCTION

Music is a universal language that transcends the barriers of language and culture. It is an essential part of the human experience, and plays an important role in many spheres of life, such as religion, culture and society. Music can move us, inspire us and unite us. It can help us express our emotions, share our experiences and build social bonds. In addition, experts in anthropology, musicology and linguistics note the essential importance that language and music have in our collective identity, from the formation of our species to the structuring of social groups such as clans, tribes, nations and cultures, as Diana Santiago argues in her book *"Musical practice, memory and language"* (2018).

1.1 PROBLEM

Significant advances have been made in the generation of music by Artificial Intelligence (AI), as evidenced by recent studies. Of particular note is the work of Su et al. (2018), who presented MuseGAN, a generative adversarial network (GAN) capable of producing high-quality music in a variety of genres, as well as the work of Yang et al. (2017), who produced MidiNet, which helps generate music in the symbolic domain, and several others, such as EmotionBox by Kaitong Zheng (2021), which produces automatic emotional music, for example.

The focus of this work is to generate music the Frevo style due to the lack of material exploring this rhythm, which is unique, and due to the lack of production on music generation in this style by AI, as well as its importance as it is a popular style and widely used in the Northeast region of Brazil, it is interesting to have experiments and studies focused on music production in this context, both at the computational research level and at the social level.

AI-based music generation has the potential to create more songs in less time, providing a more diverse and vibrant music scene. What's more, AI can personalize music to suit the individual tastes of

each listener, making it more accessible, allowing anyone to create music, regardless of musical skills or training.

Given the relevance of music, especially in the context of Frevo, a dynamic Brazilian musical genre distinguished as Intangible Cultural Heritage of Humanity by UNESCO (2012), the application of artificial intelligence (AI) in audio generation presents itself as an intriguing challenge. The faithful reproduction of distinctive elements, such as the fast rhythm, percussion instruments and vibrant melodies, emerges as a complex requirement. Understanding these complexities in the specific context of Frevo becomes crucial to fully exploiting the potential of AI in music creation.

In this challenging scenario, the research problem focuses on finding solutions that overcome the difficulties intrinsic to AI audio generation in the context of Frevo. How can we deal effectively with the faithful reproduction of such distinctive characteristics of this musical genre? This question guides the research towards innovative strategies and approaches that take into account the uniqueness of Frevo, contributing to significant advances in the intersection between music and artificial intelligence by analyzing through metrics in different areas of musical construction: Accuracy, audio quality, as as spectral and visual-spectral analysis.

Faced with the challenge of employing artificial intelligence (AI) in the generation of audio for Frevo, several questions guided the research, coming from other ideas addressed by other works on the subject of music generation and audio, taking into account the context of Frevo:

a) Reproducing the Accelerated Rhythm: Are music generation architectures capable of accurately reproducing Frevo's characteristic accelerated rhythm? This question highlights the need for innovative strategies to guarantee authenticity in the reproduction of this crucial element of the musical genre. This type of approach has already been analyzed by other studies, such as "Polyphonic Music Generation using Artificial Neural Networks" by Mendes (2019). In this study, melodies were generated using Artificial Neural Networks, taking into account

the context of polyphonic music. It doesn't develop Frevo itself, but it is a context in which the application to other styles has strong application, including Frevo.

b) Audio Quality Assessment: How can we determine the effectiveness of AI in generating audio in the context of Frevo, considering its unique characteristics? This question highlights the importance of a comprehensive evaluation approach that encompasses quantitative, subjective and spectral aspects. Frevo's fast rhythmic cadence, melodic variations and distinctive cultural elements demand an evaluation that goes beyond purely technical metrics. Therefore, in addition to precision metrics such as MAE, MSE, RMSE and signal quality such as SNR and PSNR, for example, it is essential to incorporate visual-spectral evaluations that capture the human perception of the authenticity and expressiveness of the audio generated. In addition, spectral metrics can offer insights into the fidelity of the reproduction of Frevo's distinctive acoustic characteristics. This multi-faceted approach not only enriches understanding of the performance of AI models, but also reflects the complexity and richness of Frevo's musical context, contributing to the development of more contextually relevant and effective models

c) Visual-Spectral Analysis: How can visual-spectral analysis complement the analysis of audio generation in the context of Frevo? The incorporation of visual-spectral analysis offers an additional perspective in assessing the fidelity and authenticity of generated musical productions. By examining characteristics such as the distribution of energy along the frequency spectrum and the presence of visual patterns that reflect intrinsic aspects of Frevo, such as the brightness and quality of the rhythm, we can gain valuable insights into the perceptual quality of the generated audio, this is the idea defended by Wyse (2017).

1.1.1 General objective

The main objective of this research is to evaluate the application of audio generation techniques based on artificial intelligence (AI). By specifically addressing the

In the context of Frevo, a rich and fast-paced musical genre, the aim is to develop innovative strategies and improve analytical methods, incorporating specific metrics and considering the visual-spectral perception generated. This approach aims not only to achieve advanced technical results in AI audio generation, but also to contribute to enriching the intersection between music and intelligent technologies, promoting broader advances in the field of artificial intelligence applied to music production.

1.1.2 Specific objectives

The specific objectives of this work are:

a) Bibliographic Survey: Carry out a comprehensive review of the literature related to the application of artificial intelligence in audio generation, with a focus on the techniques used to reproduce specific characteristics of Frevo. This includes investigating neural network architectures, audio processing algorithms and quality assessment methods;

b) Mapping Suitable Metrics: Identifying and selecting suitable metrics to assess the fidelity and authenticity of AI-generated audio in the context of Frevo. This involves not only accuracy metrics such as MAE, MSE, RMSE and signal quality metrics such as SNR and PSNR, but also spectral metrics and visual-spectral evaluation to capture relevant perceptual and loudness aspects. This is the understanding of the study *"Quality Enhancement for Highly Degraded Music Using Deep Learning-Based Prediction for Lost Frequencies"* by Serra and others (2021);

c) Evaluation of State-of-the-Art Methods: Analyze and compare state-of-the-art methods in AI audio generation, with a focus on their ability to reproduce the fast rhythm, percussion instruments and melodies characteristic of Frevo. This includes experiments using different neural network architectures, such as LSTM, WaveGAN, CNN, Autoencoders and GRU, to determine which perform best in generating Frevo-style audio;

d) Visual-Spectral Evaluation: Perform visual-spectral evaluations to understand the visual perception of the different types of spectrograms generated from the different neural networks and make a visual comparison of what was generated in one network and what was different in another, such as wave size, space and time generated, etc. This type of evaluation was used in the work *"Audio spectrogram representations for processing with convolutional neural networks"* by Wyse (2017).

These specific objectives are essential to achieving the central aim of the research, which is to evaluate the applications audio generation techniques based on artificial intelligence in the context of Frevo. By carrying out each stage methodically and comprehensively, it is hoped to make a significant contribution to advancing knowledge at the intersection of music and artificial intelligence, promoting the development of more contextually relevant and effective models for generating musical audio.

1.2 STRUCTURE OF DISSERTATION

This work is structured as follows: Chapter 1 presents the contextualization of the work, the problem to be studied, the questions that support the reason for this research, as well as the general and specific objectives for this study. Chapter 2 provides a theoretical background, describing in more detail Machine Learning in the context of music generation and the main types of Machine Learning, as well as the neural networks and metrics used to validate this research. In Chapter 3, work related to the topic of music generation with Artificial Intelligence is explored, showing other cases in which the most different types of neural networks are used in different contexts, but with the main focus on music generation. Chapter 4 presents the Method with the Proposal, the assembly and configuration of the entire test execution environment for this research.

Chapter 5 shows the experiments and the results themselves, highlighting the Evaluations of Precision, Qualitative, Spectral and Visual-Spectral in the

musical production in the Frevo style in each of the different types of neural networks proposed and discussed in Chapter 2. The most assertive conclusions are also presented in this chapter. In Chapter 6, the Final Conclusion is presented, with the limitations found in this research and thoughts on future work that could be done. Some suggestions are put forward as possible research topics so that researchers can have a better reference of what and how to better structure their respective research.

2 THEORETICAL BACKGROUND

to Nguyen (2019), Machine Learning is the application of statistics to computer models that make predictions based on a previously provided database.

This approach to the ability of computers to learn from experience and improve their performance automatically is also advocated by Michie, Spiegelhalter and Taylor (1994), and has been applied in various areas, including music generation. Faul (2019) expands on this definition by describing learning as a complex phenomenon that involves discovering new knowledge through observation and experimentation. In this context, machine learning in automatic music generation can be understood as the research and digitization of music learning processes at different levels of representations.

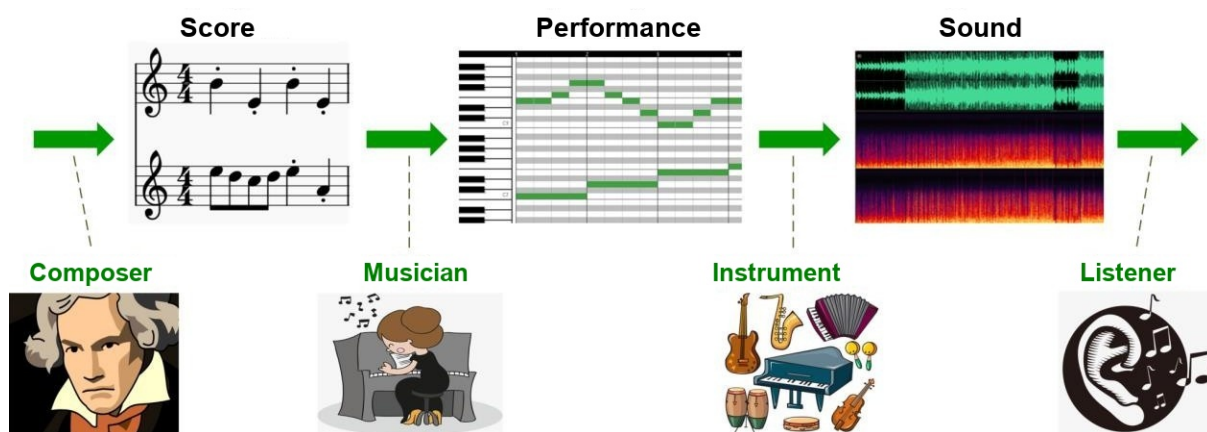
Now, by exploring automatic music generation, we can understand how the principles of machine learning are applied to create musical compositions autonomously and innovatively. Let's take a closer look at how these principles intertwine to shape the landscape of music today.

2.1 MUSIC GENERATION AUTOMATIC

About two hundred years ago, the renowned American poet Henry Wadsworth Longfellow supposedly stated: "Music is the universal language of mankind", an idea that received validation from researchers at Harvard University in 2019, led by scientist Samuel Mehr - a member of the Harvard Data Science Initiative. According to the paper "*A Comprehensive Survey on Deep Music Generation: Multi-Level Representations, Algorithms, Evaluations, and Future Directions*" by Shulei Ji and others (2020), the correlation between musical notation and its auditory manifestation mirrors the relationship between written text and spoken language. Musical notation functions as a visual representation

notation is a deeply symbolic and abstract form capable of preserving and effectively communicating musical ideas, while the auditory experience encompasses a continuous and tangible signal that encodes every discernible detail. These two forms can be delineated on distinct levels, with notation occupying the top tier and sound residing below. The interpretation and expression of music predominantly depends on performance nuances; for example, altering the tempo of a lively composition can give it a melancholic tone. Consequently, an intermediate extract can be introduced between the upper and lower layers to delineate performance complexities. Thus, the process of generating music generally unfolds in three stages, as illustrated in Figure_01_english: during the initial stage, composers draw scores; in the subsequent phase, performers execute these scores, bringing the music to life; finally, in the final stage, the performance undergoes sonification, with varied timbres (instruments) superimposed, and is perceived by the human audience.

Figure_01_english - Music



Source: Ji, Luo and Yang, 2020, p. 2.

In the contemporary music scene, artificial intelligence (AI) plays a significant role, especially in audio generation. A notable example of this application is the context of Frevo, an energetic Brazilian musical genre recognized by UNESCO (2012) as Intangible Cultural Heritage of Humanity. However, generating audio for Frevo using AI presents complex challenges, requiring the accurate reproduction of characteristic elements such as the fast-paced rhythm, percussion instruments and vibrant melodies.

In this challenging scenario, several studies have contributed to significant advances. The work *"MuseGAN: A Musical Generative Adversarial Network"* (Su et al., 2018) stands out by presenting MuseGAN, a model based on Generative Adversarial Networks (GANs) capable of generating high-quality music in different genres. Metrics such as Mean Squared Difference (MSE), Signal to Noise Ratio (SNR) and Normalized Signal to Noise Ratio (PSNR) were used to evaluate the quality of productions. Furthermore, in *"A Comparative Study on Variational Autoencoders and Generative Adversarial Networks"* (Sami, 2019), the comparison between two machine learning techniques for generating data including music - Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs) - reveals valuable insights. The results suggest that GANs are better suited to applications that demand originality and creativity, while VAEs excel in scenarios that require consistency with a training dataset.

In parallel, the article *"MidiNet: A Convolutional Generative Adversarial Network for Symbolic-domain Music Generation"* (Yang et al., 2017) introduces a specific Convolutional Generative Adversarial Neural Network (GAN) for generating music in the symbolic domain. Trained on a dataset of MIDI melodies, MidiNet demonstrates the ability to create new and original melodies. These studies not only reflect the diversity of approaches at the intersection of music and AI, but also indicate a continuous evolution in this promising field.

2.2 FREVO

Since our focus is on generating audio time series in the Frevo rhythm, let's take a brief look at the history to put it into context. According to Diniz (2012), the origin of frevo is still a controversial subject. Some scholars believe that frevo emerged at the end of the 19th century as a mixture of African and European rhythms. Others believe that frevo has older origins, dating back to the colonial period. Frevo is a fast-paced and contagious rhythm. Frevo music is usually played by frevo bands, which use instruments such as trombones and saxophones,

trumpets and drums. Frevo dancing is characterized by fast, acrobatic steps performed by the revellers.

Frevo dancing is a form of bodily expression that combines fast, acrobatic movements with traditional capoeira steps. The revelers wear colorful adornments, such as hats, scarves and umbrellas, which are used to create visual effects. Frevo is one of the symbols of Brazilian popular culture. It is a living cultural manifestation that is passed down from generation to generation. Frevo is practiced on various occasions, such as Carnival, popular festivals and cultural events. According to UNESCO, frevo was inscribed on the Representative List of the Intangible Cultural Heritage of Humanity in 2012. The inscription recognizes the importance of frevo as a Brazilian cultural manifestation.

2.3 TYPES OF MACHINE LEARNING

The field of Machine Learning, a branch of Artificial Intelligence, has risen to prominence due to the increase in computing power and the availability of vast data sets. As Latah and others (2016) note, it uses data available in its environment to learn from experience, and is used to improve performance. There are three types of Machine Learning approach: Supervised Learning, Unsupervised Learning and Reinforcement Learning. The main characteristics each are set out below.

Unsupervised Learning, a relevant facet of Machine Learning, stands out for discovering hidden patterns without human intervention, and is widely applied in sectors such as marketing, customer segmentation and recommendation systems, as well as clustering modeling and dimensional reduction (Ghahramani, 2003). Supervised Learning, in turn, according to Carmo (2013), is a class of algorithms within machine learning that has the characteristic of inferring a function given a set of data that is called the training set. Through this algorithm, the values will be corrected and their function adapted to the expected value in the output.

On the other hand, Reinforcement Learning, another crucial area of Machine Learning, focuses on an agent's ability to learn to make decisions in dynamic environments. Xu (2020) broadens this perspective, indicating that reinforcement learning encompasses prediction and control tasks, being crucial for estimating policy values and finding optimal policies that maximize expected rewards.

2.3.1 *Deep Learning* - a technique for implementing Artificial Intelligence

An algorithm-like approach to the early Machine Learning movement, Artificial Neural Networks have come and gone time. Neural networks are a way of representing an understanding of the human brain - interconnections between neurons (perceptrons, in the case of a neural network). However, unlike a biological brain where any neuron can connect to any other neuron within a certain physical distance, these artificial neural networks have discrete layers, connections and directions of data propagation and ways in which this data will be processed.

There is still no real consensus on the definition of Deep Learning. It is a repertoire of Machine Learning techniques based on artificial neural networks. The main aspect that everyone agrees on is the term deep, i.e. there are many layers processing many levels of hierarchical abstraction, which are automatically extracted from the data (Briot; Jean-Pierre, 2020).

Applications that use *Deep Learning* have performed a wide variety of tasks within *Machine Learning*, such as: classification, prediction and, recently, translation. For each type of task, there is a better approach and type of neural network that best suits the reality of this neural network. The main types of existing neural networks are: ANN (*Artificial Neural Network*), CNN (*Convolutional Neural Networks*) and RNN (*Recurrent Neural Network*), and they also have subtypes.

2.4 MUSIC GENERATION BASED ON MACHINE LEARNING

Artificial Intelligence (AI) has the potential to revolutionize the music industry. McFee (2019), in a recent study, identified two approaches

The first is the rule-based approach, which uses a set of rules that can be derived from music theory, common usage patterns or the user's personal preferences. The first is the rule-based approach, which uses a set of rules, which can be derived from music theory, common usage patterns or the user's personal preferences. The second is the machine learning-based approach, in which an algorithm learns to generate music from example data, and can be trained on existing music or created from scratch.

McFee (2019) also identified challenges to be overcome if AI-based music generation is to reach its full potential. These challenges include the need to develop algorithms that can create truly original music, address issues of control by giving users the ability to specify style and other characteristics, and consider ethical aspects such as the potential for plagiarism and the exploitation of human musicians.

Despite these obstacles, AI-based music generation has the potential to be a powerful tool for creating new and exciting music. As AI technology continues to develop, we can expect to see even more advances in AI-driven music in the future.

2.5 SUPERVISED APPROACHES

Supervised learning is a class of algorithms within machine learning that has the characteristic of inferring a function given a set of data called the training set. This set is given by the input-output pair where the input has the characteristics of the example itself and the output the desired value. Using this set of data, the algorithm will correct itself and adapt its function to the expected output value (Carmo, 2013, p. 25).

This type of approach is characterized by the use of previously labeled data sets to train algorithms, which perform tasks such as classifying data or predicting results with high precision. During the training process, as the input data is fed into the model, its weightings are iteratively adjusted until the model achieves an adequate fit, a process that is often part of cross-validation. Supervised learning plays a crucial role in solving a variety of large-scale real-world problems,

exemplified, for example, by classifying emails as *spam* and directing them to a separate folder in the inbox.

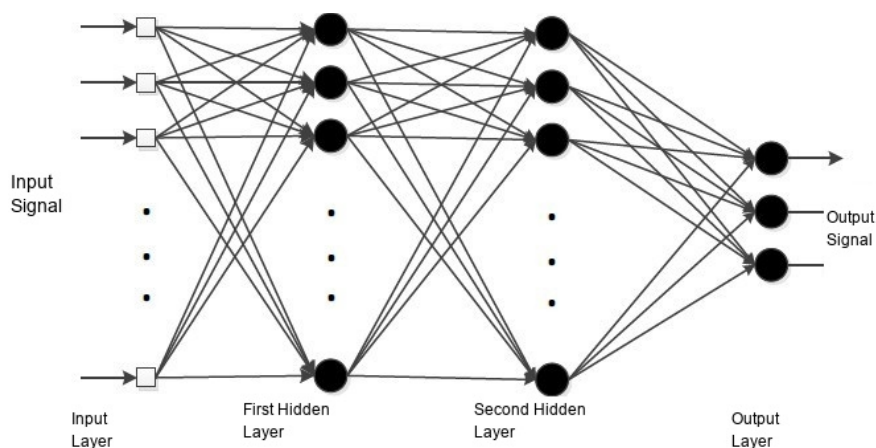
2.5.1 LSTM (*Long Short-Term Memory*) networks

LSTM networks are a type of RNN (*Recurrent Neural Network*) with modifications to the structure of its layers. In an LSTM neural network, the aim is to propagate the input at time t , as well as the state of the network in the previous step through the use of structures called gates.

Specifically, an LSTM cell contains three types of control gates:

- a) gate to decide which information is not relevant, the *Forget Gate*;
- b) gate to define which inputs will be used to update the memory cells, the *Input Gate*;
- c) gate that decides what the outputs of the cell will be, taking into account the input and the state of the memory cell, the *Output Gate* (Brownlee, 2017).

Figure_02_english - Multilayer Artificial

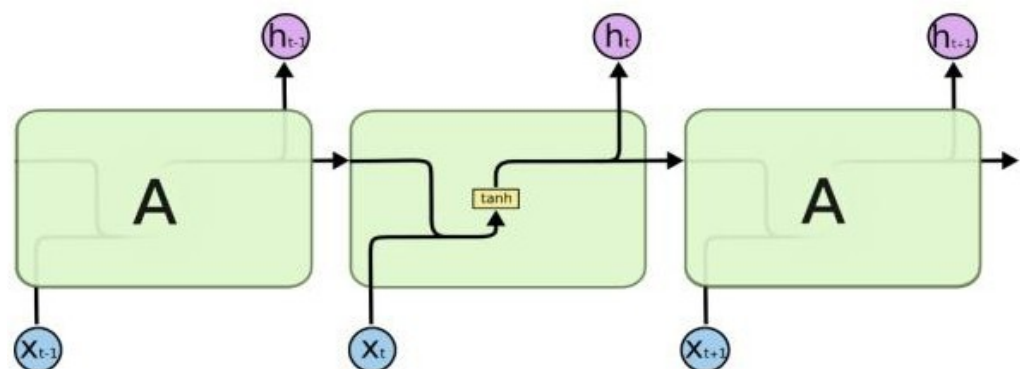


Source: Do Carmo, 2013, p. 37.

Each of the connections in a neural network has a weight associated with it. A perceptron can weight an input highly if it has a significant impact. As a neural network is trained, it is already aware of what its *output* will be after training the model. The actual *output* and the *loss functions* score how well the network performed. Loss functions can be the expected and actual absolute difference or the root mean square error of many other functions. The neural network will try to minimize the losses by adjusting the weights of the connections through a callback process. ANN interactions involve the *inputs* being calculated and passed through the perceptrons to the final *outputs*, which is used in a loss function with the expected result, and finally the weights are adjusted through back propagation. A further enhancement to the ANN is the *Long Short-Term Memory* (LSTM) network. This type of network, shown in Figure 2, allows a model to consider what has occurred previously as well as a normal set of inputs. The LSTM network will play a fundamental role in this project, not least because musical notes have a logical relationship with the sequence of notes that preceded it.

According to Nguyen (2019) and shown in Figure_03_english, this type of a model to take into account what has happened previously, as well as the normal set of inputs. This is because in the LSTM network, musical notes are logically linked to the notes that precede them.

Figure_03_english - Diagram of a *Long Short-Term*



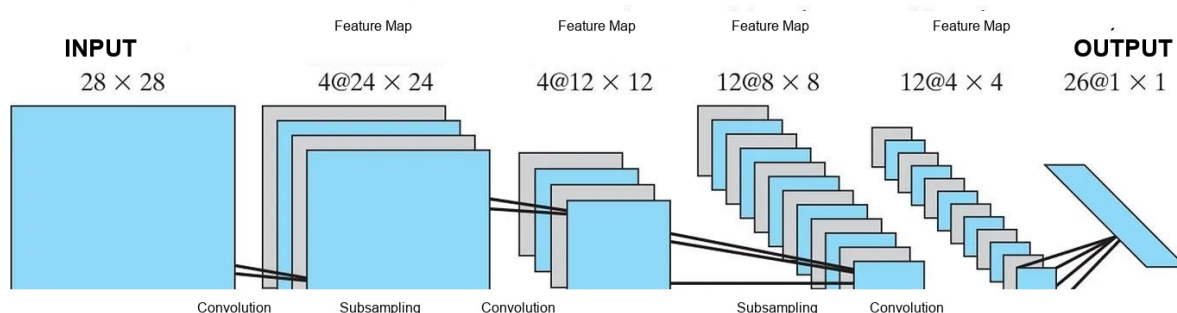
Source: Nguyen, 2019.

In this case, we have an input $\mathbf{x}_t$, for example, which passes through an entire circuit to then generate a state $\mathbf{h}_t$. Then there is a previous state, but what generates the next state is not simply the same layer feeding back.

2.5.2 CNN networks (*Convolutional Neural Network*)

Convolutional neural networks (CNNs) are a type of artificial neural network that is widely used in computer vision tasks. CNNs are inspired by the structure of the human brain and are able to learn representations of visual data that are relevant to the task at hand, as depicted in

Figure_04_english - Convolutional Neural



Source: Image extracted, edited and translated from Lecun (1998).

An important work on CNNs is "*CNN Music Emotion Classification*", by Xin Liu and others (2017). This work introduced a method that combines music spectrograms with a CNN to predict emotional *tags*. Despite the performance gain, there is room for adjustment in the CNN architecture and in the selection music segments. In addition, it was concluded that the lack of understanding of the CNN outputs hinders interpretations of the emotions inspired by the music.

Convolutional Neural Networks (CNNs) excel in computer vision tasks such as image classification, object detection and image segmentation. In addition to their effectiveness in these areas, their relative ease of training through supervised learning techniques, allowing the learning of mappings of inputs to desired outputs, contributes to their popularity. The scalability of CNNs is remarkable, making possible to train them on large data sets. In the context of applications, CNNs play key roles in several areas, such as face recognition, object identification, image segmentation, image generation, natural language processing and voice recognition, consolidating themselves as a versatile and powerful technology for solving a wide range of problems.

2.5.3 GRU networks (*Gated Recurrent Unit*)

Gated Recurrent Units (GRUs) represent a class of recurrent units in neural networks, notable for their efficiency compared to traditional units such as *Long Short-Term Memory* (LSTM). Composed of two ports, the update and reset ports, GRUs offer control over retention and discarding of information from the previous layer in the current output, represented in Figure

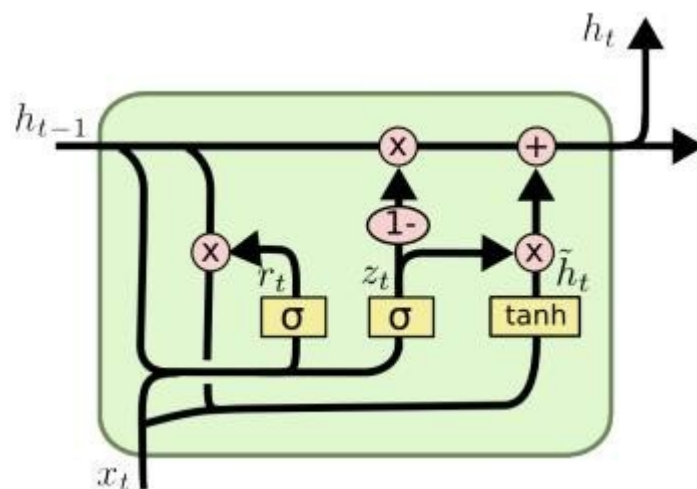
5. Their versatility makes them applicable in a variety of areas, from natural language processing to pattern recognition and robotics tasks.

In the article "*Empirical Evaluation of Gated Recurrent Neural Networks of Sequence Modeling*", by the authors Chung and others (2014), attention was focused on more advanced units that use the gating mechanism, such as the Long-Term Memory Unit (LSTM) and the *Gated Recurrent Unit* (GRU). The performance of these units in polyphonic music modeling and speech signal modeling tasks was investigated. The results showed that these more advanced recurrent units did indeed outperform more traditional recurrent units, such as *tanh* units.

Another example highlighted comes from research carried out by Rajib Rana and others (2016) comparing the effectiveness GRU and LSTM in classifying emotions from noisy speech. Although LSTM performs slightly better in certain scenarios, GRU stands out for its shorter execution time. This relationship between GRU's accuracy and time complexity is especially advantageous for embedded applications.

In summary, GRUs are emerging as a powerful tool in a variety of applications, being more efficient and more easily trainable than conventional recurring units.

Figure_05_english - GRU (*Gated Recurrent Unit*)



Source: Abdulwahab et al, 2017.

GRUs (*Gated Recurrent Units*) are a recurrent neural network (RNN) architecture used to model data sequences. They consist of two main components: the *reset* gate and the *update* gate. The *reset* gate controls the amount of information from the previous state that is retained in the current state, while the *update* gate determines the amount of information from the current input that is incorporated into the current state.

The symbols used include r_t for the reset gate, z_t for the update gate, h_t for the candidate output state and $\hat{h}_t$ for the actual output state at time t . Element-to-element multiplication is represented by $\odot$ and the sigmoid function is represented by σ . It also denotes the concatenation of vectors. The parameters include weight matrices (W_r , W_z , W_h) and bias vectors (b_r , b_z , b_h) for the gates and the output. The final states of the direct and inverse GRU units are denoted by h_{fn} and h_{bn} , respectively.

The explanation details how the candidate output state is calculated using the sigmoid function and the weights of the W_h matrix. The actual output state is obtained by multiplying the candidate output state by the update gate and adding the bias b_h . Finally, the final states of the direct and inverse GRU units are concatenated to form the *tweet* input.

2.6 NON SUPERVISED APPROACHES

Unsupervised learning is a machine learning technique in which users do not need to supervise the model. Instead, it allows the model to work on its own to discover patterns and information that was not previously detected, as it deals with unlabeled data. Labeled data could be described as a set of data that has an identification, a "label" for the observations (Araújo, 2021, p. 15).

This type of approach is characterized by the use of unlabeled data sets to train algorithms, which perform tasks such as pattern discovery, dimensionality reduction and data generation. During the training process, unsupervised algorithms seek to identify patterns in the data without the need for specific guidance.

2.6.1 *Autoencoder* networks

Autoencoders are a type of artificial neural network used to learn efficient encodings of unlabeled data. An *autoencoder* learns two functions: an encoding function that transforms the input data and a

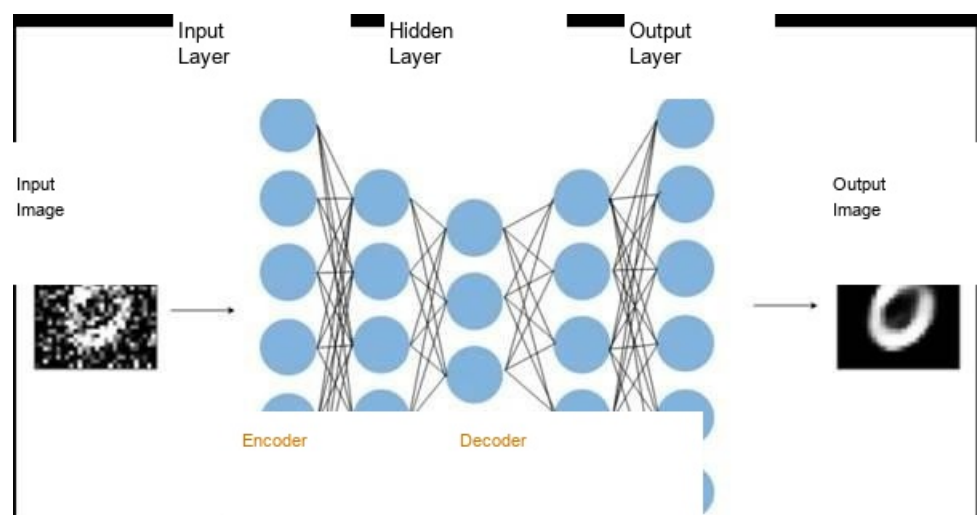
decoding function that recreates the input data of the coded representation.

The concept of *autoencoders* was introduced by Bernard Widrow and his colleagues in 1988. The original article, "*Adaptive Boltzmann Machines for Speech Recognition*" (*IEEE Transactions on Acoustics, Speech and Signal Processing*, 1988), is considered one of the pioneering works in machine learning.

The basic architecture of an *autoencoder* is made up of two main layers: an input layer, an encoding layer and a decoding layer. The input layer receives the input data, which can be images, audio or text. The coding layer transforms the input data into a smaller coded representation. The decoding layer recreates the input data in the coded representation.

Training an *autoencoder* is an iterative process. At the start of training, the *autoencoder* is randomly initialized. Input data is then fed into the *autoencoder* and the output of the decoding layer is compared with the original input data. The *autoencoder* is then updated to minimize the difference between the output of the decoding layer and the original input data, as shown in

Figure_06_english - Graphical representation of an



Source: The Keras Blog, 2024.

2.7 APPROACHES BY REINFORCEMENT

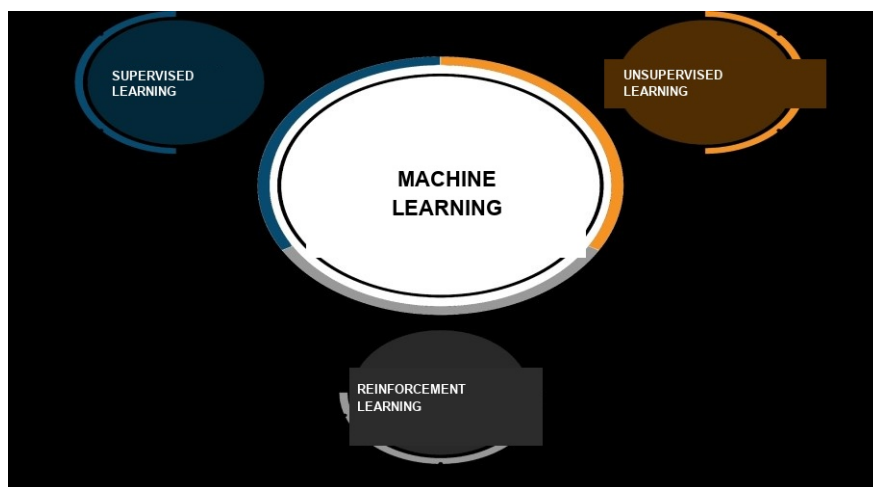
According to Sutton (2018), reinforcement learning is a type of machine learning in which the agent learns to make decisions in a dynamic environment. The agent receives rewards or punishments for its actions, and learns to make decisions that maximize the rewards.

This type of approach is characterized by the use of an agent that interacts with an environment to learn how to make decisions. The agent receives rewards or punishments for its actions, and learns to make decisions that maximize the rewards. In control tasks, the agent is not given a policy. The agent's goal is to find the optimal policy, which is the policy that maximizes the expected reward.

Xu (2020) states that reinforcement learning algorithms can be seen as a way of transforming unfeasible dynamic programming methods into practical algorithms. Dynamic programming is a method for solving optimization problems in a dynamic environment. However, dynamic programming methods can be infeasible for large-scale problems.

Figure_07_english shows a summary of the main Machine Learning this research.

Figure_07_english - Diagram of the types of learning in



Source: Dataat, GitHub, 2024.

2.7.1 GAN Networks (*Generative Adversarial Network*)

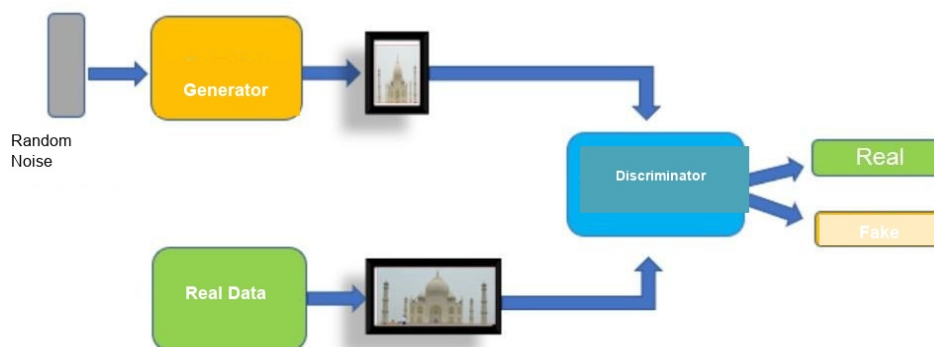
GANs, or generative adversarial neural networks, are a type of machine learning model that consists of two models, a generator and a discriminator. The generator is responsible for generating new data and the discriminator is responsible for distinguishing between real and generated data.

The concept of GANs was introduced by Ian Goodfellow and his colleagues in 2014. The original article, "*Generative Adversarial Networks*", is considered one of the most influential papers in machine learning. Training a GAN is an iterative process. At the start of training, the generator generates random data. The discriminator then tries to distinguish between the generated data and the real data. The generator is then updated to generate data that is more difficult for the discriminator to distinguish. The discriminator is also updated to better distinguish between the generated data and the real data.

Over the course of training, the generator learns to generate data that is increasingly indistinguishable from the real data. The discriminator also learns to distinguish better and better between the real and generated data.

GAN architecture:

Figure_08_english - Diagram of a GAN (*Generative*



Source: Lashgari et al., 2020, p. 108885.

In this research we will be using the WaveGAN architecture. WaveGANs are a variation of GANs that are specifically designed to generate audio data.

WaveGANs use a WaveNet-based generator to generate high-quality audio data.

3 JOBS RELATED

Initially, we did a search on *Google Scholar* using keywords in Portuguese and English, such as "*music generation with intelligence*", "*music generation with artificial intelligence*" etc., and some relevant work has been done and is still being done, such as Soundraw (2021) developed by the company Soundraw, which is an online platform that enables the creation of music with artificial intelligence, offering tools to generate melodies, harmonies, rhythms and instrumentation.

Another more recent app is Kompoz (2022) from the company Kompoz, which is an app that uses artificial intelligence to help users compose music, offering suggestions for melodies, harmonies and rhythms. As no more detailed data was found in scientific articles on these works and some others, we will not reproduce more in-depth details about them in this study.

The work "*ML-Mandolin*" (Nguyen, 2019) was developed to classify songs of the same style and from there, generate new songs in the Vietnamese style using the mandolin instrument through an LSTM network. This work was the text and basic experiment for this current work where we are developing ML-Frevo, with significant customizations and differences, such as output on only 1 channel in ML-Mandolin and a reduced number of songs compared to ML-Frevo.

The study "*MuseGAN: A Musical Generative Adversarial Network*" (Su et al., 2018) stands out for presenting MuseGAN, a model that uses Generative Adversarial Networks (GANs) to create high-quality music in various genres. Metrics such as Mean Squared Difference (MSE), Signal to Noise Ratio (SNR) and Normalized Signal to Noise Ratio (PSNR) were used to assess the quality of the productions.

Furthermore, in "*Comparative Study on Variational Autoencoders and Generative Adversarial Networks*" (Sami, 2019), a comparison was made between

two machine learning techniques for generating musical data: Variational Autoencoders (VAEs) and Generative Adversarial Networks (GANs). The results indicate that GANs are more suitable for applications that require originality and creativity, while VAEs excel in scenarios that require consistency with a training data set.

In parallel, the article "*MidiNet: A Convolutional Generative Adversarial Network for Symbolic-domain Music Generation*" (Yang et al., 2017) presents MidiNet, a Neural Neural Generative Generative Adversarial (GAN) developed specifically to generate music in the symbolic domain. Trained with a dataset of MIDI melodies, MidiNet demonstrates its ability to create new and original melodies. These studies not only demonstrate the diversity of approaches at the intersection of music and artificial intelligence, but also indicate continued progress in this promising field.

In the work "*EmotionBox: a music-element-driven emotional music generation system using Recurrent Neural Network*" by Kaitong Zheng and others (2021), *EmotionBox* is presented, an automatic emotional music generator driven by musical elements. Unlike previous methods already proposed, this model dispenses with musical data sets labeled with emotions. Instead, it uses the density of notes and the histogram of their pitch to represent the degree of arousal and the valence of the music, respectively. From this, various combinations of arousal and valence are mapped to different emotions, according to the "Russell emotion model".

The experimental results revealed that the method used in the *EmotionBox* work has a more competitive and balanced performance in generating emotional music than methods based on emotion labels. In particular, better performance was tested in the generation of music with low arousal. It was concluded that, based on note density and pitch histogram, the method was able to generate engaging and expressive music, and could be targeted to a specific emotion.

In the article "*A Comprehensive Survey on Deep Music Generation*", Ji and others (2020) propose to provide a comprehensive overview of deep learning techniques used in music generation. They classify music generation tasks into three levels: scores, performance and direct audio. For each level, they discuss different deep learning algorithms used, such as recurrent neural networks (RNNs), convolutional neural networks (CNNs), autoencoders and adversarial neural networks. adversarial generative (GANs). They also discuss different musical representations suitable for each task. To assess the quality of the music generated, they discuss subjective and objective methods.

The authors conclude that *Deep Learning* music generation is a promising area with the potential to democratize music creation and lead to new forms of musical expression.

The metrics used were:

- a) **Listening tests:** human listeners evaluate the quality of the music generated;
- b) **Feedback analysis:** human listeners provide feedback on the generated music;
- c) **Similarity metrics:** measure the similarity between the generated music and human music;
- d) **Structure metrics:** measure the structure of the music generated;
- e) **Content metrics:** measure the content of the music generated.

In summary, the article provides a comprehensive overview of deep learning techniques used in music generation. The paper discusses different algorithms, representations and evaluation methods, and points to promising future directions.

In the paper "*MusPy: A Toolkit for Symbolic Music Generation*" (Hao-Wen et al., 2020), the authors present MusPy, which is a Python library

for music generation that offers tools for data management, pre-processing and model evaluation. The article presents statistical analyses of eleven ensembles supported by MusPy, including generalization experiments between ensembles.

The results offer insights into domain overlap between the datasets and indicate that some contain more representative cross-genre samples than others, providing guidance for the choice of datasets in future research. In short, MusPy is presented as an essential tool for the development of systems for music generation, with statistical analyses and experiments that help in the choice of datasets and in improving the generalization of machine learning models when combining heterogeneous sets.

These works differ in relation to the focus of the study proposed in this work, because although they used neural networks to generate music, which is also used in this study, they did not focus on generating music in the Frevo style, as this is a totally Brazilian style, as previously discussed.

It is also worth noting that in this study, the approach is to produce new music using these neural networks that have been used in several other studies, including those mentioned in this paper, but the customization and hyperparameters that are being used come from a ready-made base of LSTM networks that is from 2019 and that since then, no new technique has been implemented for Frevo at the research level to find out which neural network could provide the best results for this purpose, specifically.

The table below presents a summary of what was covered in the related works and how they compare with this work in relation to musical production in the Frevo style, specifically:

Table 1- Summaries of the works mentioned

Aspect	EmotionBox (2021)	Ji et al (2020)	MusPy (Hao-Wen et al, 2020)	MidiNET (2017)	MuseGAN (2018)	ML-Mandolin (2019)	ML-Frevo (2024)
Application	Generation of Emotional Music	Music Generation	Music Generation	Generation of music in different formats	Generate music from various types of instruments from scratch.	Vietnamese Music Generation at Mandolin	Frevo Music Generation
Application of Neural Networks	GRU	RNN, CNN, Autoencoders, GAN	Transformer, CNN	CNN and GAN	GAN (generator and discriminator)	LSTM	LSTM, WaveGAN, CNN, Autoencoder, GRU
Evaluation Evaluation	Quality judgment	Listening tests, feedback analysis, similarity metrics structure and content	Automatic subjective evaluation in melodic composition	Quantitative and qualitative assessment, as well as additional assessments such as interactive experimentation	Human evaluation, loss of discriminatory and loss of loyalty	Classification musical style	Analysis of precision, qualitative, visual-spectral and spectral
Results	More competitive and balanced in generating of emotional music	Promising area with the potential to democratize music creation	Comparable to humans in melodic composition and harmonic accompaniment	Promising performance in the creation of symbolic music, with the ability to create complex music that is highly faithful to the desired musical style	Capable of generating music of high quality music and also accompany generated music	Musical generation of audio time series in the style of Vietnamese music	Final analysis of the best neural network options for musical composition in the Frevo style based on the data collected
Differences	It uses density histogram of height to represent excitement and musical valence	Overview of deep learning techniques for music generation	Tools for pre processing and evaluation of models, statistical analysis of data sets	CNN- GAN architecture becoming an innovative conditioning mechanism able to generate music in different formats	Architecture multi-track generating and accompanying symbolic music	Focus on generating musical time series in the style of Vietnamese music and classifying the audio in the musical style before feeding the dataset with data for training	Focus on the generation of Frevo-style music, specific evaluation for this style and verification of the metrics for this purpose

Source: prepared by the author, 2024.

To explain further, the differences highlighted in the table are in relation to the main aspects of each work mentioned.

Application: Refers to the main purpose of applying music generation techniques. While *EmotionBox* aims to generate emotional music, Ji et al focus on music generation in general, MusPy provides a tool for generating music with various functionalities, *ML-Mandolin* focuses on generating music from various types of instruments from scratch, MuseGAN excels at generating music in different formats, and MidiNet concentrates on generating music in the symbolic domain.

Neural Network Architecture: Indicates the types of neural networks used in each work. While *EmotionBox* uses GRU (*Gated Recurrent Unit*), Ji et al employ a variety of architectures, such as RNNs, CNNs, *Autoencoders* and GANs. MusPy uses *Transformer* and CNN. ML-Mandolin uses CNN and GAN, MuseGAN uses GAN (generator and discriminator), and MidiNet uses LSTM, WaveGAN, CNN, Autoencoder and GRU.

Evaluation Method: This refers to the methods used to evaluate the quality of the generated songs. While *EmotionBox* uses quality judgment based on note density and pitch histogram, Ji et al (2020) employ listening tests, feedback analysis, similarity metrics, structure and content. *MusPy* uses automatic subjective evaluation in melodic composition. *ML-Mandolin* uses human evaluation, discriminator loss and fidelity loss, MuseGAN uses Mean Squared Difference (MSE), Signal to Noise Ratio (SNR) and Signal to Normalized Noise Ratio (PSNR), and *MidiNet* uses quantitative and qualitative evaluation, as well as additional evaluations such as interactive experimentation.

Results: Indicates the results achieved by each work. While *EmotionBox* achieves competitive and balanced results in the generation of emotional music, Ji et al point to the promising democratization of music creation, *MusPy* achieves results comparable to those of humans in melodic composition and harmonic accompaniment, *ML-Mandolin* is capable of generating high quality music.

MuseGAN stands out in its ability to generate music in different formats, and *MidiNet* achieves promising results in the creation of symbolic music, with the ability to create complex songs that are highly faithful to the desired musical style.

These differences are important to highlight the specific approaches each work and how they differ in their objectives, methods and results.

4 METHOD

This chapter presents the Proposal, Material, *Setup* and Method adopted to carry out this research.

4.1 PROPOSAL

In this research, with the focus on generating and evaluating audio time series in the Frevo style, in a first step for musical generation, we will load the data and limit it to a specific interval. Next, we will create training and test sets by dividing the temporal data into sequences. Subsequently, we will train the models using the training set and generate predictions for the sets separated for this purpose. After that, we'll save the predictions and the original music part. It's important to note that the data will always run on 2 channels for *stereo* sound rather than mono.

This research comes from the basis of the work done by Nguyen (2019) in his work "*Machine Learning Mandolin*", in which he creates Vietnamese music using LSTM networks, but he does not explore the metrics data, only the form of classification and architecture for creation, is used only from LSTM. In this study, the metrics are explored from the source code of the work he used for *Mandolin*, using a customization for Frevo adding more songs and exploring the metrics where he didn't explore. The other neural networks were based on analysis of other data generation work with audio in general, not just music, and the metrics were audio metrics. It's also worth remembering that the architectures used LSTM as the music extraction model, etc., since this process is the same for all the neural networks, but the training itself on the different neural networks was customized for testing and using the same or similar hyperparameters, so that the study could have consistent data for comparison.

The metrics used to assess quality, prediction and the audio time series are *Mean Absolute Error* (MAE) and *Mean Squared Error* (MSE),

Root Mean Squared Error (RMSE), *Signal-to-Noise Ratio (SNR)* and *Peak Signal-Noise Ratio (PSNR)* for each channel and globally. It also calculates spectral metrics such as spectral centroid, spectral bandwidth and spectral contrast for each channel.

It displays spectrogram graphs for visual assessment of the quality of the generated music compared to the original. Calculates Sound Pressure Level (SPL) and Signal-to-Noise and Distortion Ratio (SNDR) metrics for each channel and globally.

4.2 EVALUATION METRICS

The most common assessment of the accuracy of a neural network model predicting future stock values is based on computing the model's error. The lower the error of a model, the better adapted it is to the data set and the closer its prediction is to the actual values. This assertion is supported by recent research in its results, such as the work of Steurer et al (2020).

In this section, the metrics that will be used in this work will be presented for comprehensive evaluation of the audio time series generated.

4.2.1 MAE - Mean Absolute Error

It is an absolute difference metric. This metric limits the space of outliers in model performance and is, however, highly efficient where there are quality problems in the data set (Steurer et al., 2020). It is a simple and intuitive measure of accuracy that is less sensitive to outliers than other metrics, such as the mean square error (MSE). The lower the error value, the closer to the expected results. The equation below expresses the MAE formula:

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_{-i} - \hat{y}_{-i}|$$

Where:

- n: total number of observations in the dataset
- Σ : this symbol denotes the sum. The MAE formula calculates the sum of absolute differences between actual values and predictions for all observations in the dataset.
- $\hat{y}_{-i}$: is the predicted or expected value of the i-th observation
- y_{-i} : actual value of the i-th observation
- $| * |$: is the absolute value

4.2.2 MSE - Mean Squared Error

This is the mean square error, which is a heavier measure of error than the MAE. The lower the MSE, the better the forecast accuracy. The mean squared error (MSE) is an evaluation metric that measures the average squared difference between the actual values and the predicted values. According to Steurer et al. (2020), this metric is extremely sensitive to extreme points and is useful in situations where it is crucial to minimize significant forecast errors.

$$MSE = \frac{1}{n} \sum_{i=1}^n (y_{-i} - \hat{y}_{-i})^2$$

Where:

- n: is the total number of samples in the dataset
- y_{-i} : is the actual value of the i-th observation
- $\hat{y}_{-i}$: is the predicted or estimated value of the i-th observation
- Σ : sigma, mathematical notation representing the sum
- $(y_{-i} - \hat{y}_{-i})^2$: is the difference between the actual and predicted value for the i-th sample, squared to ensure the result is always positive
- $1/n$: is the inverse of the total number of samples, turning the sum of squared differences into an average, making the metric independent of dataset size

4.2.3 RMSE - Root Mean Squared Error

RMSE - Root *Mean* Squared Error is the square root of the MSE, which is a more intuitive measure of error than the MSE. The lower the RMSE, the better the forecast accuracy, according to Steurer et al (2023).

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_n - \hat{y}_i)^2}$$

Where:

- n: total number of observations or samples in the dataset
- y_n : is the actual (or observed) value of the i-th sample
- $\hat{y}_i$: represents the predicted (or estimated) value of the i-th sample by the model
- Σ : sigma, summation symbol
- $(y_n - \hat{y}_i)^2$: is the difference between actual and predicted values for the i-th sample, squared to ensure the result is always positive
- $1/n$: is the inverse of the total number of samples, transforming the sum of squared differences into an average, making the metric independent of dataset size
- $\sqrt{}$: square root of the expression inside the parentheses

4.2.4 SNR - Signal-to-Noise Ratio

The signal-to-noise ratio (SNR) is an evaluation metric that measures the ratio between signal energy and noise energy. It is a measure of signal quality, expressed in decibels (dB). It is a fundamental metric for assessing the quality of a signal, the higher the better. This is what the work *"Using audio quality to predict word error rate in an Automatic Speech Recognition System"* by Fish and others (2006) makes clear.

$$SNR = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise}} \right)$$

Where:

- P_{signal} : Signal power is the power of the signal of interest
- P_{noise} : Noise power is the power of the noise present in the signal
- $\log_{10}$: is the base-10 logarithm
- 10: is a conversion factor to the decibel (dB) scale

4.2.5 PSNR - Peak Signal-to-Noise Ratio

This is the peak signal-to-noise ratio, which measures the difference between the maximum values of the signal and the noise. The higher the PSNR, the better the signal quality. This is demonstrated by Ichigaya and others (2006) in their article "*A method of estimating code PSNR code using quantizing DCT coefficients*".

$$PSNR = 10 \log_{10} \left(\frac{peak\ signal\ value^2}{MSE} \right)$$

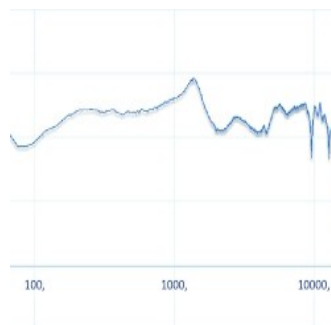
Where:

- $peak\ signal\ value^2$: The peak signal value is the maximum value that can be represented by the signal, squared.
- MSE: Is the Mean Squared Error, the average of squared differences in amplitude between the original and processed/compressed audio signal samples
- $\log_{10}$: is the base-10 logarithm
- 10: is a conversion factor to the decibel (dB) scale

4.2.6 SPL - Sound Pressure Level

Sound pressure level (SPL) is an evaluation metric that measures intensity of sound. It is expressed in decibels (dB), where 120 dB is the limit of human hearing without pain or discomfort, i.e. the SPL can be considered good.

Figure_09_english - Graph of sound pressure level
function of frequency in Hz



Source: Cunha, Julia and others (2016), p. 954.

SPL is calculated using the following formula:

$$SPL = 20 \log_{10} \left(\frac{p}{p_{ref}} \right)$$

Where:

- p : is the sound pressure of the sound
- p_{ref} : is the reference pressure, generally defined as 20 μPa
- $\log_{10}$: is the base-10 logarithm
- 20: is a conversion factor to transform the pressure ratio into decibels (dB)

4.2.7 SNDR - Signal-to-Noise-Distortion Ratio

The signal-to-noise-distortion ratio (SNDR) is an evaluation metric that measures the ratio between signal energy, noise energy and distortion energy. It is a measure of signal quality, which is expressed in decibels (dB).

$$SNDR = 10 \log_{10} \left(\frac{P_{signal}}{P_{noise} + P_{distortion}} \right)$$

Where:

- P_{signal} : is the power of the signal of interest
- P_{noise} : is the noise power.
- $P_{distortion}$: is the distortion power.
- $\log_{10}$: is the base-10 logarithm
- 10: is a conversion factor to the decibel (dB) scale

4.2.8 Visual-Spectral Assessment

It carries out a comparative visual-spectral evaluation between the original audio and the audio generated by the models tested. Rather than relying solely on objective metrics such as mean absolute error (MAE) or mean square error (MSE), etc., visual-spectral evaluation involves human perception. In this case, it is an approach to assessing the perceptual quality of the generated audio compared to the original audio. Rather than relying solely on objective metrics such as mean absolute error (MAE) or mean square error (MSE) etc., visual-spectral evaluation involves human perception.

In the specific case of the project, it uses spectrogram graphs to visualize the spectral characteristics of the original audio and the generated audio. These graphs help visually identify similarities and differences between the two audio signals. The spectrogram shows how the intensity of the sound varies at different frequencies over time. By comparing the spectrograms of the

and the generated audio, it is possible to observe important patterns and characteristics, such as the presence of specific frequencies, changes in intensity over time and other spectral properties. This visual analysis allows for a more intuitive assessment of the quality of the generated audio compared to the original audio.

In summary, visual-spectral evaluation seeks to capture the human perception of audio quality, providing a more holistic and intuitive perspective, complementing traditional objective metrics. This approach is valuable, especially when the goal is to create models that generate audio with a perceived quality similar to the original audio.

4.2.9 Visual-Spectral Assessment

Objective spectral metrics are calculated for a specific audio sequence. Here some more specific spectral metrics are used to calculate these values for a specific audio sequence.

The metrics calculated are:

- a) ***Spectral Centroid***: The center of mass of the spectrum;
- b) ***Spectral Bandwidth***: The effective bandwidth of the spectrum;
- c) ***Spectral Contrast***: A measure of the difference between peaks and valleys in the spectrum.

The MAE, MSE, RMSE, PSNR and SNDR metrics are all suitable for assessing the accuracy of an audio time series forecast. The SNR and SPL metrics are less commonly used to assess the accuracy of audio time series predictions. SNR measures the ratio of signal to noise, while SPL measures the sound pressure level of the signal. These metrics are more often used to assess audio quality, rather than prediction accuracy. Both functions are used to subjectively and objectively evaluate the quality of the generated audio compared to the original audio, providing both a visualization and numerical metrics.

So, we can separate everything discussed as follows, as shown in Table 2.

Table 2 - Metrics for Evaluating Accuracy and Audio Quality

Category	Metric	Description
Precision	MAE	Mean Absolute Error
Precision	MSE	Mean Square Error
Precision	RMSE	Square Root of Mean Squared Error
Signal Quality	SNR	Signal-to-noise ratio
Signal Quality	PSNR	Peak Signal-to-Noise Ratio
Audio Quality	SPL	Sound Pressure Level (Average)
Audio Quality	Spectral Centroid	Average Spectral Center
Audio Quality	Spectral Bandwidth	Average Spectral Bandwidth
Audio Quality	Spectral Contrast	Average Spectral Contrast

Source: prepared by the author, 2024.

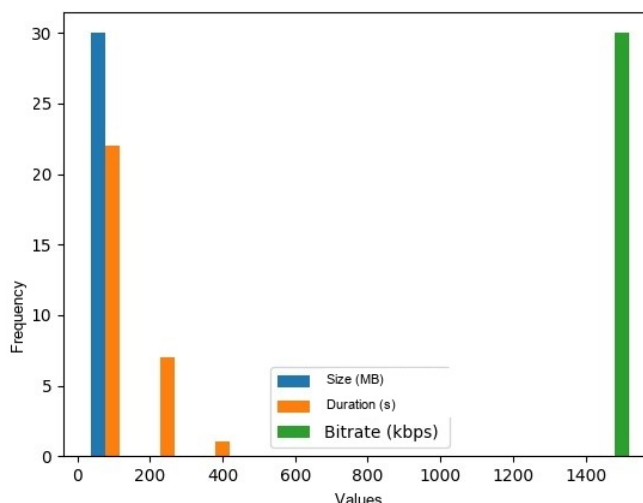
The SPL metric was added to the audio quality category because it is a measure of the intensity of sound. This measurement is important for audio quality, as louder sounds are generally more pleasant to listen to within a decibel band fit for the human ear.

4.3 MATERIALS

Initially, 30 instrumental songs in the Frevo (street) style by different artists were downloaded from a *YouTube* playlist using the *aTube Catcher* program, which is freely available on the internet. The 30 songs selected were based on the focus of the work, which was to generate music in the Frevo style and, above all, the style performed in the streets during the Carnivals of Recife/PE. The number of songs was decided by placing approximately one third more songs than the original work *Mandolin*, which initially had 19 songs. The files were saved in .mp3 format with high sound quality, and the .mp3 files are saved compressed to reduce their size.

An online converter was used to transform the .mp3 file into .wav. This is the data format that will be used by the machine learning model for the tests in this study. To prepare the data, 30 Frevo songs were collected from a *YouTube playlist*, each lasting between 2min20s and 3min30s. They were placed in a folder called Frevo. The songs were read and used from this folder in all the scripts for all the neural network tests (Fig. 10).

Figure_10_english - Histogram of Song Size, Duration Averages



Source: prepared by the author, 2024.

Analyzing the data, the data below was obtained by running a *script* that is also available to check in the repository:

Average file size (MB): 30.23

Average song length: 2 minutes 45 seconds Average song

bitrate (kbps): 1531.84

4.4 EXECUTION ENVIRONMENT AND EXPERIMENTS

An application was written in Python 3.11 running on a machine running *Windows 11 PRO 64-bit*, 2.60 GHz Core i7-13650 HX 13th Gen Intel® Core(TM) processor with 32GB RAM and using IntelliJ IDEA 2023.3.2 (*Ultimate Edition*) from JetBrains. Remember that any IDE can be used in this case.

The neural networks used in this research are: LSTM (*Long Short Term Memory*), GAN (*Generative Adversarial Network*), *Autoencoders*, CNN (*Convolutional Neural Networks*) and GRU (*Gated Recurrent Unit*). The neural networks chosen were LSTM - because as mentioned, this study is a continuation of the previous study "*ML-Mandolin*" by Nguyen (2019), WaveGAN, CNN, *Autoencoders* and GRU - because most of the studies done where music generation by neural networks is the focus, used these networks as the basis of their research.

The hyperparameters used to test the generation of audio time series will be the same or close to the values tested in most of them, based on the LSTM as the initial reference parameter. This approach will provide a more in-depth analysis of the behavior of the different neural networks in a given specific context, considering the same hyperparameters with the same or approximate values. This method will enable a more accurate and meaningful comparison between the models, providing valuable *insights* into their relative performances and potential advantages or limitations in relation to the hyperparameter settings.

The hyperparameters used in each of the neural networks were the same or with some specific change due to the architecture of the neural network, but in a specific way, as shown in Table 3.

Table 3 - Table with common hyperparameter data

Hyperparameter	LSTM	CNN	GRU
Lot size	50	50	50
Times	50	50	50
Input format	(1, 100)	-	(None, 1, 100)
Activation of dense layers	Linear	ReLU	Leaky ReLU
Loss function	MSE	MSE	MSE
Optimizer	Adam	Adam	Adam

Source: prepared by the author, 2024.

Table 4 - Table with specific hyperparameter data

Hyperparameter	WaveGAN	Autoencoders
Sequence size	100	-
Number of interactions	10000	500

Window size	25	-
Filter pitch	4	-
Resource size	100	-
Learning rate (discriminator)	Adam	-
Learning rate (generator)	Adam	-
BatchNormalization Momentum	0.8	-
Filter size	-	3
Number of filters	-	64
Learning rate	-	0.0001
Number of units per layer	-	256, 128, 64, 32, 64, 128, 256

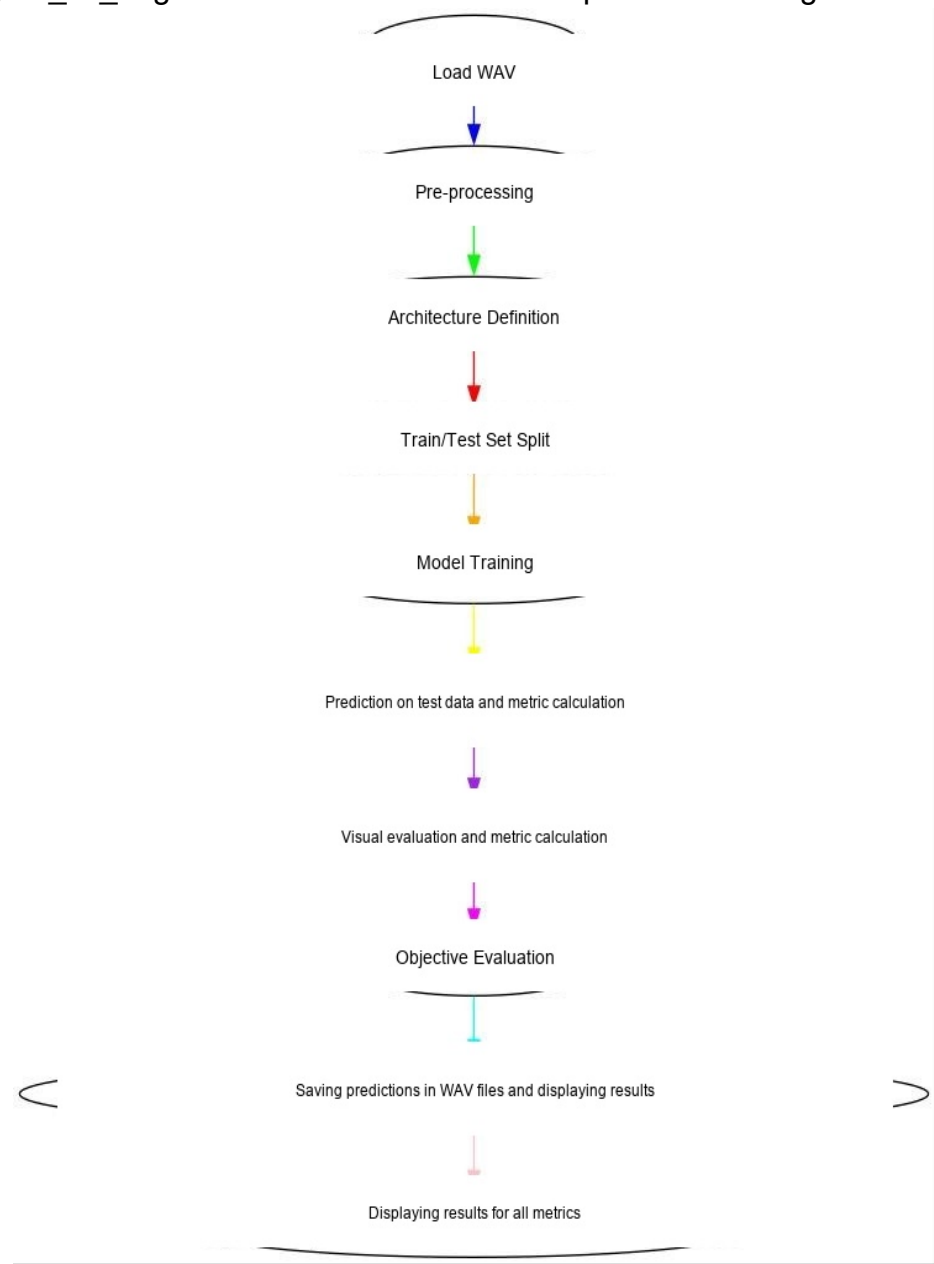
Source: prepared by the author, 2024.

This data represents the hyperparameters used in each model to test the generation of audio time series, providing a comparative basis for analyzing the behavior of the different neural networks. It is worth remembering that WaveGAN does not have a "*shape*" parameter like the other models, as its architecture is based on convolutional networks that operate directly on the real input sequence.

5 EXPERIMENTS AND RESULTS

When the program is run, the flow followed by the five neural networks tested (LSTM, WaveGAN, CNN, *Autoencoder* and GRU) is as shown in

Figure_11_english - Flow followed for audio production using neural



Source: prepared by the author, 2024.

The program loads the 30 WAV audio files and performs some pre-processing operations on the data, such as splitting the training and test sets, creating data sets with sliding windows to feed the specific neural network and reshaping the data into the appropriate format.

After that, he defines the architecture of the neural network, compiles and trains two separate models, one for each audio channel - we're always working with 2 audio channels since the sound is in stereo format.

After training the neural network, the program makes predictions on the test data using the trained models and calculates various error metrics of the generated audio in relation to the original audio, i.e. the test audio in relation to the training audio. In these calculations, various metrics are used, such as: MAE (Mean Absolute Error), MSE (Mean Square Error), RMSE (Root Mean Square Error), SNR (Signal to Noise Ratio) and PSNR (Peak Signal to Noise Ratio). It also calculates spectral metrics such as spectral centroid, spectral bandwidth and spectral contrast for each audio channel and globally.

In addition, a visual-spectral evaluation is carried out, printing the spectrograms of the original and generated audio for human evaluation. It also calculates SPL (Sound Pressure Level) and SNDR (Signal to Noise Ratio and Distortion) for objective evaluation of the generated audio.

Finally, the generated predictions are saved in WAV files and display the results of all the calculated metrics, providing a comprehensive analysis of the model's performance in generating music and its comparison with the original audio.

5.1 ACCURACY ASSESSMENT AND QUALITATIVE

Metric	LSTM	WaveGAN	CNN	Autoencoders	GRU
MAE	526.11	0.05	436.84	663.49	511.75
MSE	847879.37	847879.37	360776.93	1466633.62	794021.75
RMSE	920.67	920.67	600.55	1211.04	890.70

Metric	LSTM	WaveGAN	CNN	Autoencoders	GRU
SNR(dB)	-5.28	-5.28	-55.56	-6.66	-5.89
PSNR (dB)	-1.15	-1.15	-7.43	-1.33	-1.86

Metric	LSTM	WaveGAN	CNN	Autoencoders	GRU
SPL(dB)	-328	-328	-761	150.77	-315
SNDR(dB)	27.83	27.83	26.97	-954	28.12

According to Bishop (2006), in relation to Audio Accuracy metrics, the lower the MAE, MSE and RMSE, the better the accuracy of the model. The lowest Mean Absolute Error (MAE) value was for WaveGAN, so it was more accurate in this context, obtaining a very low value of only 0.05. It can be seen that the CNN model stands out for its better accuracy, as evidenced by the lower Mean Square Error (MSE) and Root Mean Square Error (RMSE) values, denoting a remarkable predictive capacity. However, it is important to note that this accuracy is attenuated by the lower signal quality expressed by the lower Signal-to-Noise Ratio (SNR) and Peak Signal-to-Noise Ratio (PSRN) indices, which can impact the fidelity of the musical results.

On the other hand, the *Gated Recurrent Unit* (GRU) model exhibits a balance between precision and signal quality that proves particularly relevant in the production of Frevo-style music. Although it doesn't lead the way in terms of absolute precision, its ability to preserve the fidelity of the audio signal can be more highly valued in a musical context, where perceptual quality plays a crucial role. Considering the energetic nature and rhythmic complexity of Frevo, faithful reproduction of timbres and musical nuances is essential, making the GRU model a promising choice composition and music production in this style.

While the CNN model may be preferable in situations that demand absolute precision, such as critical data analysis, the GRU model offers a more balanced approach, prioritizing the perceptual quality of the audio. In this way, when employing machine learning models for musical production and composition in the Frevo style, it becomes imperative to consider not only accuracy, but also signal quality and the overall listening experience provided by each model, thus taking into account the nuances inherent in this specific context.

5.2 SPECTRAL EVALUATION (SPECTRAL METRICS GLOBAL)



Model	LSTM	WaveGAN	CNN	Autoencoders	GRU
Value	2419.27 Hz	273.75 Hz	1413.36 Hz	3799.85 Hz	2419.27 Hz



Model	LSTM	WaveGAN	CNN	Autoencoders	GRU
Value	1850.18 Hz	96.28 Hz	1619.85 Hz	3175.25 Hz	1850.18 Hz



Model	LSTM	WaveGAN	CNN	Autoencoders	GRU
Value	18.69 Hz	96.28 Hz	19.03 Hz	17.95 Hz	18.69 Hz

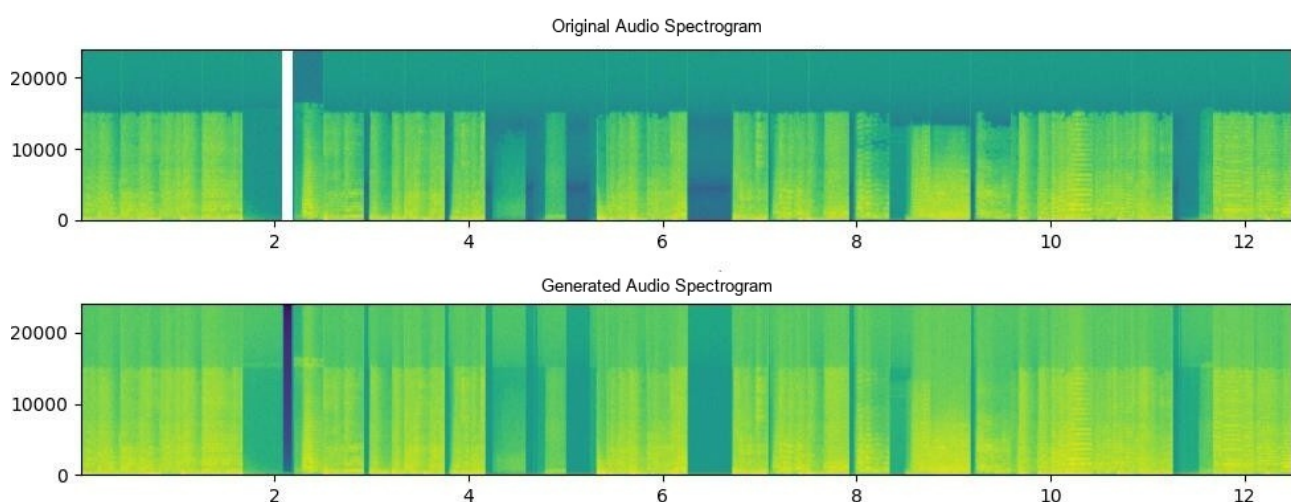
Based on the results of the global spectral metrics, WaveGAN stood out compared to the other audio generation models, such as LSTM, CNN, *Autoencoders* and GRU. Specifically, WaveGAN showed considerably higher values for both spectral centroid and spectral contrast, indicating a superior ability to capture and reproduce detailed nuances in the spectral characteristics of audio signals. For example, WaveGAN's spectral centroid was 273.75 Hz, while the other models ranged from 1413.36 Hz (CNN) to 3799.85 Hz (Autoencoders), showing a significant gap in relation to WaveGAN's performance.

According to some studies that have been done, but not conclusive, such as "*A generative model for raw audio*" by OORD et al. (2016), and "*GANSynth: Adversarial Neural Network Synthesis*" by Engel et al. (2020), they point out that minimizing spectral information can lead to improved generalization, allowing the model to capture and reproduce a wide variety of spectral features present in the training data. This results in a model that is better able to deal with different types of audio signals. In addition, minimizing spectral information can help reduce *overfitting* by avoiding excessive focus on specific details in the training data.

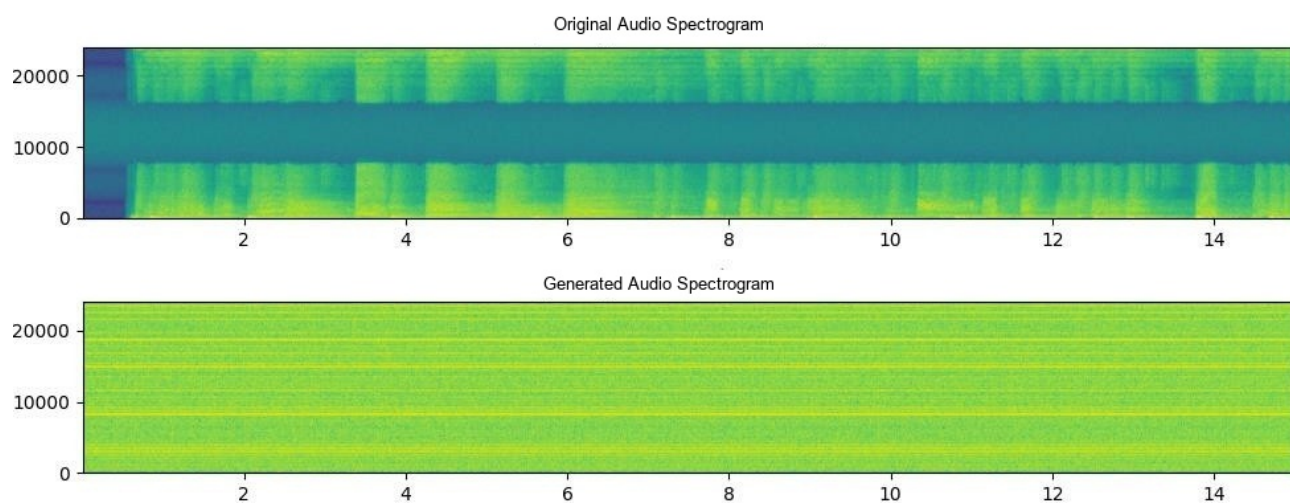
Another advantage is improved control over the quality of the audio generated. By minimizing spectral information, the model becomes less sensitive to unwanted variations in the frequency spectrum, which results in audio samples that are more realistic and pleasing to the ear. Therefore, WaveGAN obtained the best result due to its ability to minimize spectral information, which allowed for better capture and reproduction of general and relevant audio characteristics.

5.3 VISUAL ASSESSMENT- SPECTRAL

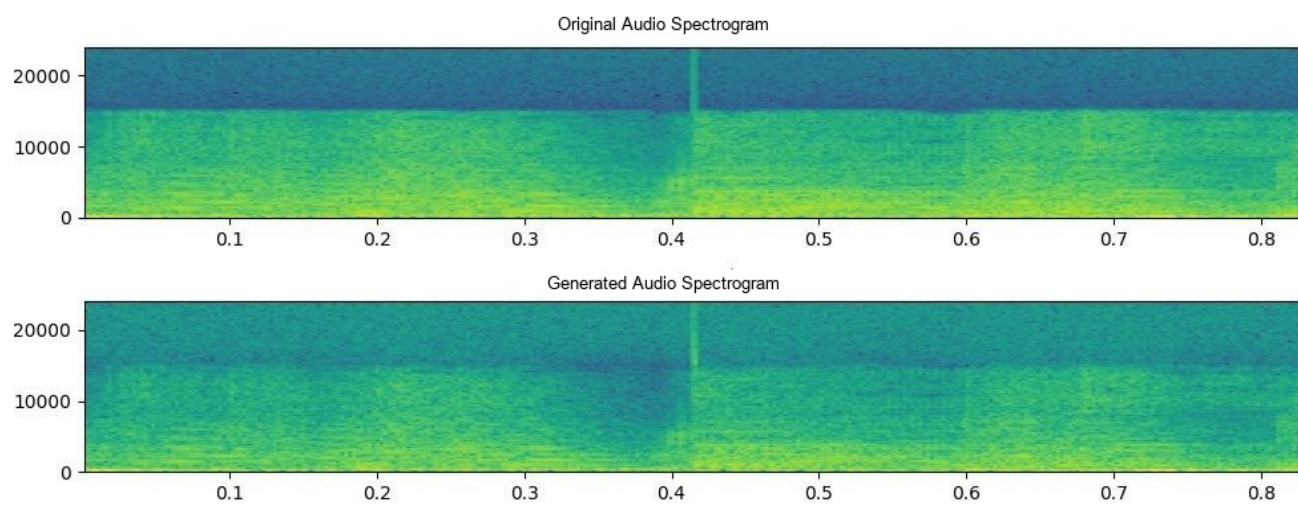
LSTM



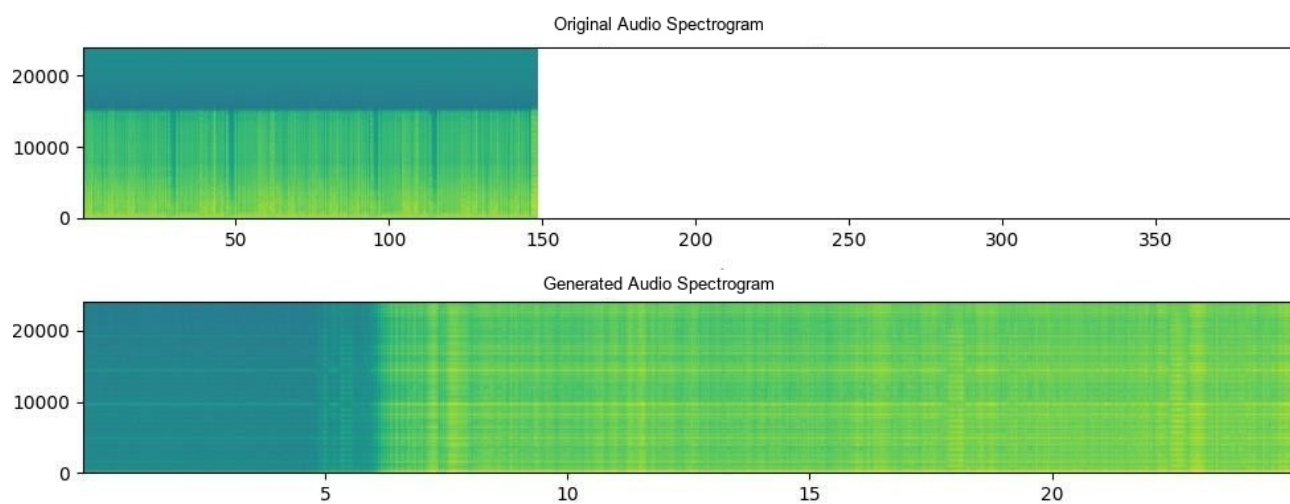
WaveGAN



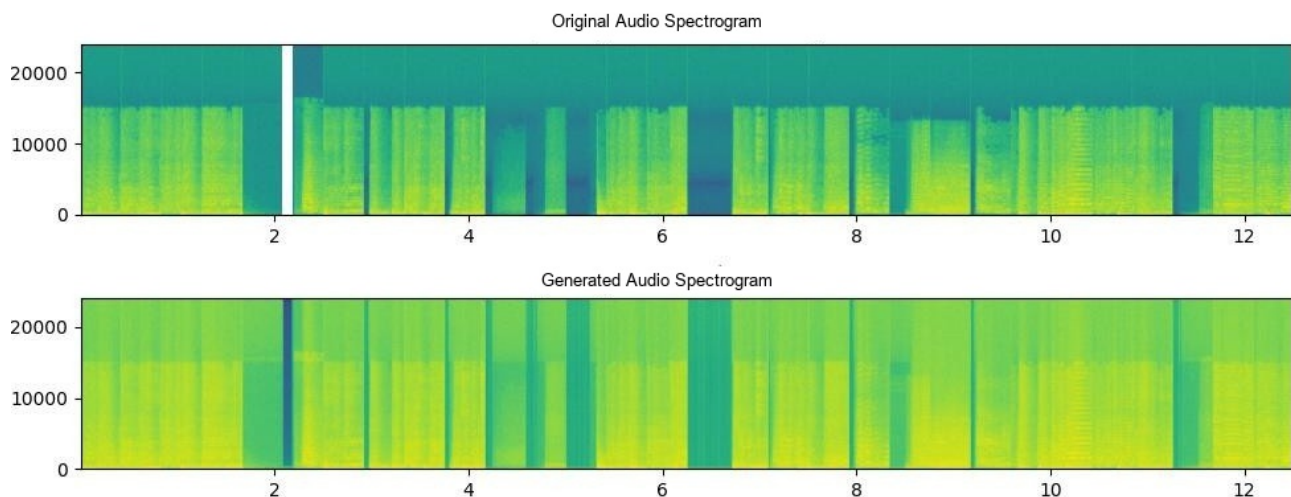
CNN



Autoencoders



GRU



Source: own elaboration, 2024.

Here, the spectrograms of these audio signals are displayed side by side to allow a visual comparison between them. Spectrograms are graphical representations that show the intensity of the different frequencies of an audio signal over time. This visualization can help in the visual-spectral evaluation of the quality of the generated audio compared to the original audio. It is a qualitative approach to evaluating aspects such as fidelity, clarity and similarity between the original and generated audio.

It can be seen that in the *Autoencoders* there is a huge difference between the original audio and the generated audio. This difference is due to the fact that of all the neural networks, it was where the audio produced was far from equivalent to the original audio, which in fact, was not a satisfactory audio with a lot of noise and hiss at the beginning and an almost imperceptible noise in the second after 2:30m of execution of the noise and hiss generating null data in the original audio spectrogram.

5.4 CONSIDERATIONS FINAL

This study analyzed some neural networks used in audio production. In the context analyzed, it was used to produce music in the Frevo style. The networks studied were: LSTM, WaveGAN, CNN,

Autoencoders and GRU. The networks use the same or very similar hyperparameters depending on their architecture so that the results can be compared later.

The results in the table show accuracy and audio quality metrics for different generation models: LSTM, WaveGAN, CNN, *Autoencoders* and GRU. It can be seen that the WaveGAN model shows superior performance in terms of MAE, indicating higher accuracy in its predictions, while CNN stands out in MSE and RMSE, suggesting lower dispersion and higher accuracy. However, all models exhibit negative SNR and PSNR values, indicating that the noise signal is more prominent than the signal of interest. Notably, the *Autoencoders* model exhibits the highest SPL, implying a higher sound intensity in its predictions, while GRU exhibits the highest SNDR, suggesting a better relationship between the signal of interest and the noise/distortion. This analysis provides a comprehensive view of the relative performance of the models on various accuracy and audio quality metrics.

The "Accuracy Metrics" and "Quality Metrics" tables provide a comprehensive analysis of the audio accuracy and quality metrics for various generation models, highlighting significant differences between them. The WaveGAN model stands out for the lowest Mean Absolute Error (MAE), suggesting superior accuracy in its predictions. In addition, CNN has the lowest Mean Square Error (MSE) and Root Mean Square Error (RMSE), indicating relatively high accuracy and a smaller dispersion in relation to the real values. However, all the models exhibit a negative Signal-to-Noise Ratio (SNR), denoting that the noise signal is more prominent than the signal of interest, with the CNN showing the lowest SNR value. As for Sound Pressure Level (SPL), the *Autoencoders* model stands out with the highest value, suggesting a higher level sound pressure in its predictions, while GRU shows the best Signal-Noise-Distortion Ratio (SNDR), indicating a better relationship between the signal of interest and the noise/distortion.

Regarding the Visual-Spectral Analysis, we can see that the graphs generated have a similarity between the LSTM, CNN and GRU graphs, indicating some kind of similarity between the generated audio and the original audio at various points in their metrics, either by accuracy or similarity. WaveGAN and *Autoencoders* clearly show that there was a dysfunction between the generated audio and the original audio, and this data is corroborated by the other qualitative, accuracy and spectral analyses.

Considering all these points discussed and the importance of perceptual quality in music production, the most suitable network for producing Frevo style music, according to the results obtained in the experiments in this study, would be the Gated Recurrent Unit (GRU) model. Although the *Autoencoder* may have slightly higher accuracy in terms of prediction, the GRU demonstrates a balance between accuracy and signal quality, which is essential for music production, especially in a context such as Frevo, where perceptual quality plays a crucial role. Furthermore, analysis of the spectrograms reveals that the *Autoencoder* model produced significantly different results from the original audio, suggesting less fidelity in reproducing the desired musical characteristics. Therefore, considering not only accuracy, but also signal quality and perceptual fidelity, the GRU model emerges as the most promising choice for music production using neural networks for the Frevo style.

In short, while each model has its strengths and weaknesses, the choice of the most suitable model will depend on the specific requirements of the project, such as prioritizing accuracy, signal quality or noise tolerance. This comparative analysis provides valuable insights to guide the selection of the most appropriate model for different audio generation applications, considering the complex interaction between the performance metrics evaluated.

6 CONCLUSION

The results obtained provide a detailed understanding the accuracy and audio quality metrics for a variety of generation models, including LSTM, WaveGAN, CNN, *Autoencoders* and GRU. Each model displays its distinctive performance, reflected in the MAE, MSE, RMSE, SNR, PSNR, SPL and SNDR metrics. The highlight is the WaveGAN model, which demonstrates superior accuracy in terms of MAE, while CNN excels in MSE and RMSE, suggesting less dispersion and greater accuracy in its forecasts. However, the analysis reveals that all the models have negative SNR and PSNR values, indicating that the noise signal is more prominent in relation to the signal of interest, suggesting the need for improvements in filtering or in the generation process to reduce the impact of noise on the results.

In addition to the metrics mentioned above, it is crucial to consider the specific context each application when evaluating the performance of the models. For example, while the *Autoencoders* model stands out with the highest SPL, indicating greater sound intensity, the GRU exhibits the highest SNDR, suggesting a better relationship between the signal of interest and noise/distortion. These nuances highlight the importance of a personalized approach when choosing the most suitable model for each usage scenario, taking into account the specific demands of the project, such as accuracy, signal quality and noise tolerance.

In addition, it is necessary to address the limitations encountered during the study, such as the hardware restrictions when running some models, mainly *Autoencoders*, which resulted in low speed and temporary unavailability of the system. Another important limitation was the scarcity of specific materials in the context of audio time series generation, which made the research more challenging and required adaptations of hyperparameters based on experiments from other areas. These limitations highlight potential areas for future improvements and refinements in audio generation methods.

Considering these results and limitations, there are several opportunities for future research aimed at improving audio generation systems. Strategies such as parameter optimization, model combination and data set diversification can further enhance model performance. In addition, exploring more advanced deep learning models, such as Generative Adversarial Networks (GANs), and investigating model interpretability techniques can provide valuable insights and drive the development of more robust and effective audio generation systems for a wide range of applications.

Directly answering the questions that guided and motivated this research:

Are music generation architectures capable of accurately reproducing the characteristic fast rhythm of Frevo? Yes, but with certain hardware limitations, hyperparameter settings, amount of data for training and testing, etc. In other words, by adjusting this data, a significant improvement in the result is expected, since this was demonstrated through the initial work of ML-Mandolin, which had 19 songs and increased the number of songs in the database by just over a third, even though it was a different style, ML-Frevo showed a better result both in LSTM, which was where ML-Mandolin performed, and in GRU, which was where ML-Frevo performed best.

How can we determine the effectiveness of AI in generating audio in the context of Frevo, considering its unique characteristics? It was demonstrated through evaluations of precision, qualitative, spectral and visual-spectral metrics, making a comparative analysis among the most different neural networks where there was gain, loss and similarity by data from these metrics as well as visual analysis. The audio generated was also made available for human analysis in the project's code repository.

How can visual-spectral analysis complement the analysis of audio generation in the context of Frevo? By incorporating the generated spectra, various visual-spectral analyses were carried out to evaluate the fidelity and

authenticity between the generated audio and the original audio. By examining the distribution of energy along the frequency spectrum, it was seen that some patterns resemble the original frevo audio, but some nuances depend on more tests and more data being collected for training.

6.1 LIMITATIONS

The results presented reveal significant opportunities to improve audio generation systems by exploring advanced parameter optimization and model combination strategies. In addition, investigating more sophisticated data pre-processing techniques and acquiring diverse data sets can enrich the analysis and provide more comprehensive results. Another promising line of research is the adoption of more advanced deep learning models, such as Generative Adversarial Networks (GANs), to explore their potential in the production of high-quality audio.

On the other hand, it is important to recognize the limitations faced during the study, such as hardware restrictions that caused slowdowns and "crashes" during the execution of certain models, especially those based on *Autoencoders*. In addition, the scarcity of specific material on generating Frevo-style audio via artificial intelligence adds complexity to the research, requiring the adaptation of techniques from other areas and experimentation with different hyperparameters. These limitations highlight the need for innovative solutions and adaptive strategies to deal with computational and data availability challenges.

Given these results and challenges, future research could focus not only on improving the performance of models, but also on their interpretability. Understanding the patterns learned by the models and integrating explainability techniques can make them more transparent and reliable, boosting their use in practical applications that require interpretability, such as medical care and forensic audio analysis, for example. This integrated approach promises significant advances in audio generation and its application in various domains.

6.2 FUTURE WORK

Considering future work, there are several opportunities to improve the results presented. It would be interesting to explore parameter optimization strategies and model combinations to further enhance the performance of audio generation systems. In addition, investigating data pre-processing techniques and acquiring more diverse data sets could enrich the analysis and provide more comprehensive results. Another promising avenue of research would be to explore more advanced deep learning models, such as Generative Adversarial Networks (GANs), to investigate their potential in generating high-quality audio. These directions could contribute significantly to the advancement of the field and to the development of more robust and effective audio generation systems in different application scenarios.

In addition to the approaches mentioned above, there is still room to explore the interpretability of audio generation models, seeking to better understand the patterns learned and how they influence the quality of predictions. Analyzing the latent representations of the models can reveal insights into the most relevant acoustic characteristics for generating high-quality audio, making it possible to identify specific areas for refining and improving the algorithms. In addition, the integration of explainability techniques can help to make the models more transparent and reliable, facilitating their adoption in practical applications where interpretability is essential, such as in medical assistance systems or forensic audio analysis, mentioned above. This holistic approach, which combines performance optimization, data diversification, advanced model exploration and interpretability, promises to significantly boost progress in audio generation and its application in a variety of domains.

Considering future work, in addition to the strategies mentioned above, it would be interesting to further explore adapting audio generation models to different types of data, such as music with specific instruments or even

even speak in different languages and accents. Diversifying data sets can help ensure the robustness and generalizability of models in a wider variety of contexts, making them more useful in real-world scenarios. Furthermore, exploring transfer learning and incremental learning techniques can be advantageous for taking knowledge acquired in one specific domain and applying it to another, reducing the need for large data sets for training. This could make it easier to adapt models to new tasks or domains with less computational effort and training time.

Alongside all this, some research has also taken an interesting direction in the area of generative AI for music, such as Google's MusicLM project, presented in the paper "*MusicLM: Generating Music from Text*" by Andrea Agostinelli and others (2023), where it creates high-quality music with 24 kHz sound fidelity that remains coherent for several minutes. *MusicLM* is able to interpret textual descriptions and transform them into pieces of music that match the setting described. To evaluate its performance, *MusicCaps* was used, an extensive database with more than 5,500 pairs of text and music, carefully selected by professional musicians. The results show that *MusicLM* outperforms other similar models, both in audio quality and in its fidelity to the input text.

Knowing this, future work could focus on generating lyrics, along with improving text conditioning and vocal quality. Another aspect is modeling the structure of songs at high levels, such as introduction, stanza and chorus. Modeling the music at a higher sampling rate is an additional goal.

Another work that is currently very interesting to highlight is *DDSP-Piano*, which was presented in the article "*DDSP-Piano: A Neural Sound Synthesizer Informed by Instrument Knowledge*" by Renault and others (2023), which is a new method of synthesizing piano sounds using deep learning. Although it achieves better sound quality than some other methods based on neural networks,

it is recognized that physical modeling techniques can still produce superior results.

Future work is that the app could benefit from the interpretability and differentiation of DDSP-Piano to handle other tasks related to polyphonic music, such as sound source separation and self-supervised multi-pitch transcription.

Ultimately, the combination of innovative approaches with a deeper understanding of models and their effects on audio generation can lead to significant advances in the field. Continued collaboration between researchers and the scientific community, together with access to increasingly powerful computing resources, offers a promising scenario for the development of more sophisticated and effective audio generation systems. The work can be downloaded free of charge from the repository: <https://github.com/lucaspiano/ml-frevo>.

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